HRTek Cloud Infrastructure & Architecture Guide

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1. Executive Summary

This document serves as the authoritative technical reference for the cloud infrastructure powering the HRTek B2B Platform. The application utilizes a **Serverless, Event-Driven Architecture** built on Amazon Web Services (AWS).

The system has recently undergone a major architectural migration—moving from a fragmented, multi-table legacy structure to a unified **Single-Table Design** to support high scalability, reduce costs, and ensure millisecond-latency data retrieval.

2. Architecture Evolution (Old vs. New)

2.1 The Legacy Architecture (Deprecated)

In the initial interactions (v1), the application used a relational style approach on DynamoDB, creating separate tables for every entity type.

- **Old Tables:**

B2b_platform
Questions
sessions.

- **Why it was replaced:**
    - **Inefficient Data Retrieval:** fetching a Company and its Employees required multiple network calls to different tables.
    - **Higher Costs:** You had to provision read/write capacity for *three* separate tables. If one spiked in traffic, the others sat idle, wasting money.
    - **Status:** These tables are now **Read-Only/Deprecated.** They are kept for archival purposes but are not used by the live application.

2.2 The Modern Architecture (v2)

The new system consolidates core business logic into a **Single-Table Design (STD).**

- **New Primary Table:**

b2b_platform

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- **Why it is used:**
    - **Performance:** We can fetch a Company, its Users, and its Orders in a single request using advanced query patterns.
    - **Cost Efficiency:** We manage capacity for only *one* table.
    - **Atomicity:** Complex actions (like a User joining a Company) happen in a single transaction, preventing data inconsistencies.

3. Database Layer (Amazon DynamoDB)

3.1 Primary Table:

b2b_platform

This is the **live** database for the application. It stores heterogeneous data (Users, Companies, Orders) in one place by using generic keys (Pk and sk).

Schema Strategy:

- **Partition Key (pk):** Represents the "Group" or "Parent" (e.g., the Company ID).
- **Sort Key (sk):** Represents the specific "Child" or "Item" (e.g., a specific User or Order).

Entity Mapping (How data is stored):

Entity	Partition Key (pk)	Sort Key (sk)	Purpose
Organization	COMPANY#<uuid>	METADATA	Stores company name, size, and settings.
User Profile	USER#<uuid>	PROFILE	Stores user email, password hash, and role.
Analysis Report	COMPANY#<uuid>	REPORT#<engine>#<id>	Stores JSON results from AI analysis.
Orders	COMPANY#<uuid>	ORDER#<uuid>	specific billing/subscription records.

Entity	Partition Key (pk)	Sort Key (sk)	Purpose
Payments	COMPANY#<uuid>	PAYMENT#<uuid>	Stores payment proof urls and verification status.
Eligibility	COMPANY#<uuid>	ELIGIBILITY#TRANSF...	Tracks assessment completion progress.

3.2 Auxiliary Tables (Special Purpose)

These tables exist outside the Single-Table Design because they have unique lifecycle requirements:

1. sessions
    - **Purpose:** Stores user login tokens.
    - **Why separate?** It uses a **TTL (Time To Live)** feature to automatically delete old sessions after 24 hours. Mixing this widely deleting data with business records in the main table is risky/complex.
2. Questions
    - **Purpose:** Stores the questions for the Engine Assessments.
    - **Why separate?** This is "Static Content." It changes rarely and is read frequently by all users. It does not belong in the transactional

b2b_platform table.

4. Storage Layer (Amazon S3)

Bucket Name:

hrtek-storage-prod

This layer handles heavy binary files that cannot be stored in the database.

Directory Structure:

- /uploads/avatars/

: Publicly accessible user profile pictures.

- /uploads/proofs/

: **Private** payment screenshots. Accessible only by verified Admins via signed URLs.

- /generated/reports/

: PDF exports of the AI analysis reports.

5. Intelligence Layer (Amazon Bedrock)

This layer powers the "Report Generation" features of the dashboard.

5.1 Model Configuration

- **Model:** DeepSeek V3 (

deepseek.model.v3

)

    - *Note:* Selected for its superior reasoning capabilities in HR/Business analytics contexts compared to standard models.

- **Region:** us-east-1(Virginia)
- **Parameters:** Temperature 0.7

(Creative but focused), Max Tokens 4096

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5.2 AI Workflow

1. **Fetch:** System pulls raw assessment answers from

b2b_platform

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2. **Construct:** A prompt is built using proprietary engineering templates.
3. **Invoke:** Request is sent to Amazon Bedrock securely.
4. **Save:** The streaming text response is captured, formatted as JSON, and saved back to

b2b_platform under REPORT#...

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6. Security & Compliance

6.1 Identity Access Management (IAM)

Strict **Least Privilege** access is enforced for the application server:

- **DynamoDB:** Allowed to

Put Item, Get Item, and Query the b2b_platform

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- **Bedrock:** Allowed only Invoke Model

.

- **S3:** Allowed to Put Object for uploads.

6.2 Data Residency

- **Primary Data (PII/Business Data):** Stored in

ap-south-1

(Mumbai) to comply with local regulations.

- **AI Processing:** Data enables transient processing in

us-east-1

(US) solely for AI inference, with no permanent storage in that region.